

CSE 471: MACHINE LEARNING

Learning Probabilistic Models

ML parameter learning (Continuous model)

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$L = \sum_{j=1}^N \log \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_j-\mu)^2}{2\sigma^2}} = N \left(-\log \sqrt{2\pi} - \log \sigma \right) - \sum_{j=1}^N \frac{(x_j - \mu)^2}{2\sigma^2}$$

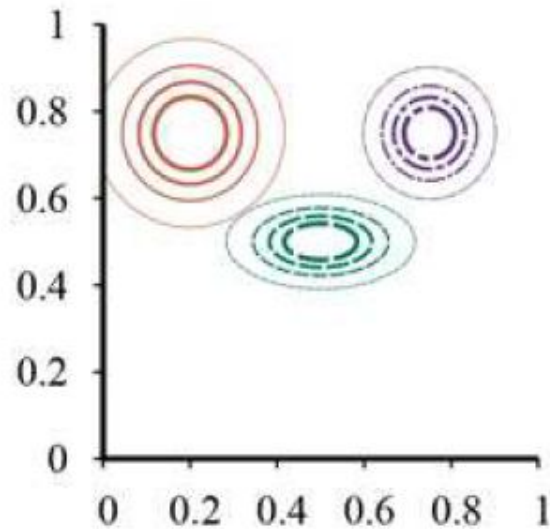
$$\frac{\partial L}{\partial \mu} = -\frac{1}{\sigma^2} \sum_{j=1}^N (x_j - \mu) = 0 \quad \Rightarrow \quad \mu = \frac{\sum_j x_j}{N}$$

$$\frac{\partial L}{\partial \sigma} = -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{j=1}^N (x_j - \mu)^2 = 0 \quad \Rightarrow \quad \sigma = \sqrt{\frac{\sum_j (x_j - \mu)^2}{N}}$$

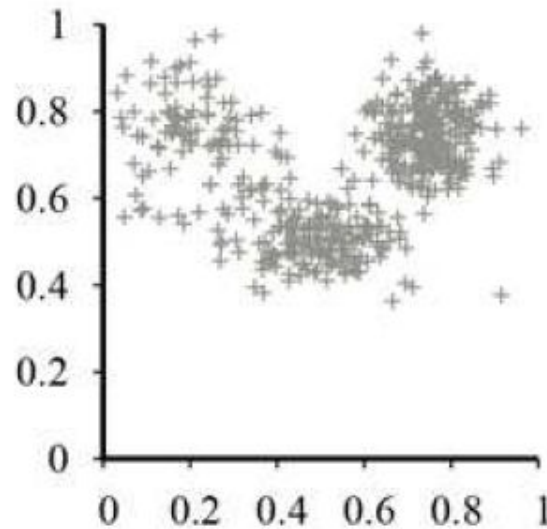


The EM Algorithm

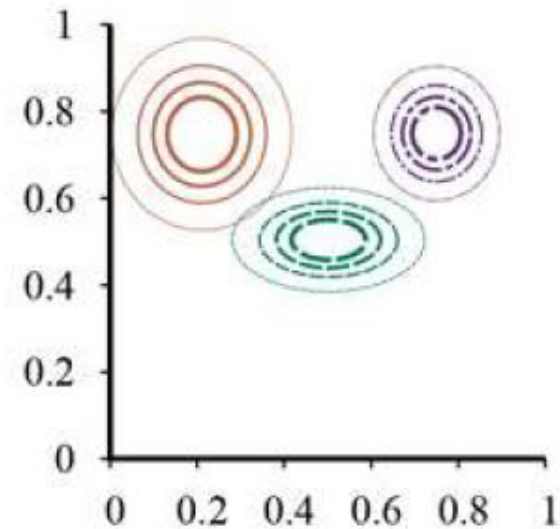
Gaussian mixture model



(a)



(b)



(c)

(a) A Gaussian mixture model with three components; the weights (left-to-right) are 0.2, 0.3, and 0.5. (b) 500 data points sampled from the model in (a). (c) The model reconstructed by EM from the data in (b).

Mixture distribution

- k components
- Each component is a distribution itself

$$P(\mathbf{x}) = \sum_{i=1}^k P(C = i) P(\mathbf{x} | C = i)$$

□ C is the random variable denoting the component

Mixture of Gaussians

- Each component is a multivariate Gaussian
- Parameters
 - The weight of each component, $w_i = P(C = i)$
 - The mean of each component, (μ)
 - The covariance of each component Σ_i

Mixture of Gaussians

- If we knew the parameters of each component
-> we could assign each data point to a component.

Gaussian Probability Density Formula

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Mixture of Gaussians

- If we knew which component generated each data point -> It would be easy to recover the component Gaussians

$$L = \sum_{j=1}^N \log \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x_j - \mu)^2}{2\sigma^2}} = N \left(-\log \sqrt{2\pi} - \log \sigma \right) - \sum_{j=1}^N \frac{(x_j - \mu)^2}{2\sigma^2}$$

$$\frac{\partial L}{\partial \mu} = -\frac{1}{\sigma^2} \sum_{j=1}^N (x_j - \mu) = 0 \quad \Rightarrow \quad \mu = \frac{\sum_j x_j}{N}$$

$$\frac{\partial L}{\partial \sigma} = -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{j=1}^N (x_j - \mu)^2 = 0 \quad \Rightarrow \quad \sigma = \sqrt{\frac{\sum_j (x_j - \mu)^2}{N}}$$

Solving by EM Algorithm

- Initialize the parameters arbitrarily
- E-Step

E-step: Compute the probabilities $p_{ij} = P(C=i | \mathbf{x}_j)$, the probability that datum $\mathbf{x}_j$ was generated by component i . By Bayes' rule, we have $p_{ij} = \alpha P(\mathbf{x}_j | C=i) P(C=i)$. The term $P(\mathbf{x}_j | C=i)$ is just the probability at $\mathbf{x}_j$ of the i th Gaussian, and the term $P(C=i)$ is just the weight parameter for the i th Gaussian. Define $n_i = \sum_j p_{ij}$, the effective number of data points currently assigned to component i .

Solving by EM Algorithm

- Initialize the parameters arbitrarily
- M-Step

M-step: Compute the new mean, covariance, and component weights using the following steps in sequence:

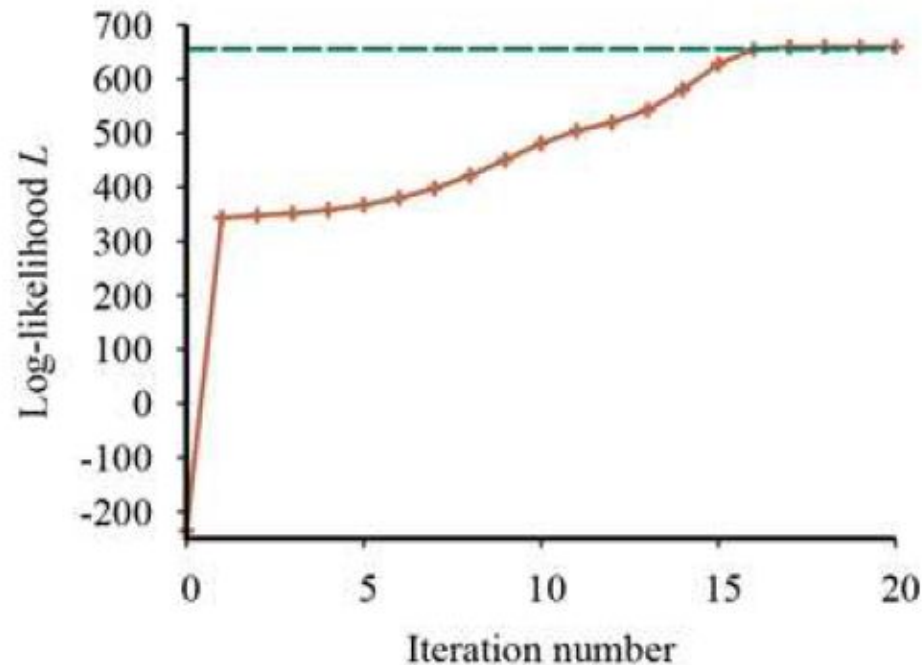
$$n_i = \sum_j p_{ij}$$

$$\mu_i \leftarrow \sum_j p_{ij} \mathbf{x}_j / n_i$$

$$\Sigma_i \leftarrow \sum_j p_{ij} (\mathbf{x}_j - \mu_i) (\mathbf{x}_j - \mu_i)^\top / n_i$$

$$w_i \leftarrow n_i / N$$

Log likelihood increasing



Graphs showing the log likelihood of the data, L , as a function of the EM iteration. The horizontal line shows the log likelihood according to the true model. Graph for the Gaussian mixture model in [Figure 20.12](#).

Saddle Point

